System Design: Kestrel R&D Formulation Assistant

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1. Problem Definition & Requirements

1.1 Context & Problem Statement

Kestrel Specialty Chemicals manufactures performance chemicals across textiles, home care, and paper. Its R&D organization comprises 8 formulation chemists operating across two physical laboratories. Today, customer-driven formulation projects require 3–4 weeks of manual desk research: searching fragmented network drives, cross-referencing raw material (RM) price sheets, conducting ad-hoc literature searches, and manually calculating active solids and stoichiometry in spreadsheets.

Kestrel requires an **AI-Assisted R&D Formulation Assistant** to compress desk-research cycle time from **weeks to days**, guiding chemists from an unconstrained customer brief to an approved initial trial plan.

1.2 Requirements Breakdown

- **Business & User Requirements:**
  - *Interactive Intake:* Transform unstructured briefs into structured targets (e.g. solids 6.0–8.0%, pH 3.5–4.5, viscosity 50–300 cP, max cost ₹22.0/kg, silicone-free, 4-week 45°C stability with no separation/yellowing, softness panel score ≥ 4/5).
  - *Evidence Retrieval:* Surface relevant historical trials and literature with explicit record-level provenance.
  - *Candidate Proposal:* Generate 2–3 candidate recipes with technical rationale and auditable cost schedules.
  - *Draft Trial Plan:* Output actionable lab protocols (equipment, addition order, mixing speeds) ready for bench execution.
  - *Chemist Governance:* **Strict human-in-the-loop authorization; zero automated physical lab bookings.**
- **Functional Requirements:**
  - *Deterministic vs. Empirical Boundary:* Enforce mass balance (100.0% ± 0.05%), unit normalization, hard constraints, and raw material costing in pure Python using `Decimal` fixed-point arithmetic—**never in LLM text generation or binary float math**. Distinguish deterministic formulation constraints from empirical properties (thermal stability, sensory softness) requiring laboratory testing. Missing/incomplete measurements must remain `UNKNOWN` or `NOT_VERIFIED`; generative LLMs must never assert empirical passes without physical lab records.
  - *Identity Resolution:* Resolve chemical synonym tokens (`TEA-esterquat`) to primary supplier codes (`RM-101` vs `RM-102`), flagging ambiguous mappings rather than silently merging them.
  - *Data Cross-Referencing:* Ingest trial data, monthly price feeds, and regulatory notes while quarantining unapproved (`T-04`), physically contradicted (`T-05`), or incomplete (`T-11`) trials.
- **Non-Functional Requirements:**
  - *Traceability:* Every suggested component percentage must trace to a specific trial ID, price feed record, or citation.
  - *Latency Target (Proposed):* End-to-end response ≤ 15s (p95 ≤ 12s; unmeasured in prototype); internal deterministic rules execute in < 50ms.
  - *Security & IP:* Strict tenant isolation; zero proprietary chemical formulations used for public LLM training.
  - *Cost Ceiling:* Departmental model API budget strictly < $1.00 per chemist/month (< $8.00/month for 8 chemists).

1.3 Scope Boundaries & Design Assumptions

- **In Scope:** Structured request parsing, candidate recipe generation, deterministic validation, cost calculation, audit logging, and chemist approval workflow.
- **Out of Scope:** Direct machine execution, automatic ERP purchase orders, autonomous instrument scheduling.
- **Core Assumption:** Historical workbooks represent noisy legacy data requiring validation barriers; regulatory notes represent internal guidance flags requiring chemist confirmation, not definitive legal conclusions.

2. System Architecture: The Modular Monolith

Rather than deploying an over-engineered multi-agent microservice architecture for 8 chemists, the platform is designed as a **Modular Monolith with Deterministic Safety Gates**. Probabilistic LLMs handle linguistic parsing and creative recipe hypothesis drafting, while deterministic Python services govern stoichiometry, unit conversions, costing, and safety enforcement.

Architectural Layer	Core Technology	Primary Responsibility
Chemist Web Workspace	React / TypeScript SPA	Brief intake, evidence inspection, recipe adjustment, role-based digital sign-off.
Orchestration Layer	FastAPI (Python 3.11+)	Workflow state machine, prompt assembly, bounded retry loops, diagnostic reporting.
Inference Gateway	Tiered Model Router	Structured JSON parsing (lightweight model) & candidate drafting (prompt-cached model).
Deterministic Barrier	Python `Decimal` Engine	Mass balance (± 0.05%), unit conversion, constraint enforcement, and price joins.
Data & Evidence Store	PostgreSQL / SQLite	Relational storage for 12 ingested workbook sheets, BM25 text index for notes/patents.
Audit & Governance	Append-Only Table	Tamper-evident session logging with SHA-256 state hashes and chemist approval tokens.

3. End-to-End Formulation Workflow

The workflow enforces a strict boundary between generative drafting and physical lab execution. Retries on deterministic failure are bounded (maximum 2 attempts) before graceful pipeline termination.

Formulation Candidate Workflow Sequence diagram · deterministic safety barrier · mandatory chemist sign-off · no automated lab booking

1-3 · INTAKE 4-5 · EVIDENCE 6-7 · GENERATE 8 · VALIDATE 9 · AUDIT 10-13 · SIGN-OFF Deterministic checks · Python Decimal engine (no LLM in this step) Mass balance 100.0% ± 0.05% · Units (cP, drum) · Active costs · Silicone ban alt *[all checks pass] [any check fails · retries < 2]* MANDATORY CHEMIST SIGN-OFF GATE · ZERO AUTOMATED LAB BOOKING **1** **Submit brief** **2** **Parse brief** → TargetSpecification **3** **Structured Target JSON** **4** **Bidirectional query** trials · prices · notes **5** **Benchmarks + failure bounds** T-05, T-04 flags **6** **Draft candidates** grounded in retrieved evidence **7** **CandidateFormulations** structured JSON **8a** **Validate candidate set** **8b** **Verified candidates + auditable costs** **8c** **Diagnostic report** failed checks + reasons **8d** **Re-prompt** max 2 retries · loops back to step 6 ↻ then repeat 6 → 7 → 8a **9** **Write tamper-evident log** evidence IDs · SHA-256 hash **10** **Present candidates** + cost schedule **11** **Adjust & sign off** **12** **Record digital signature token** **13** **Emit Approved Trial Plan** **Chemist** human reviewer **FastAPI Orchestrator** workflow control **Inference Gateway** LLM calls **Store** SQL / BM25 **Safety Barrier** deterministic **Audit Log** tamper-evident Legend Request / call Response / return Safety Barrier (deterministic) Failure / retry path Human gate Final output

Operational Workflow Stages

1. **Intake & Target Structuring:** Natural language requests are parsed into quantitative ranges: solids [6.0, 8.0]%, pH [3.5, 4.5], viscosity [50, 300] cP, max cost ₹22.0/kg, silicone prohibited, 4-week 45°C stability (zero phase separation/yellowing), and softness panel score ≥ 4/5.
2. **Bidirectional Evidence Retrieval:** Simultaneously queries passing project benchmarks (`TX-SOFT-26`) and failed/contradictory trials (`T-05` phase separation at Week 3; `T-11` missing data) with defect causes attached, preventing the model from re-exploring documented chemical failures.
3. **Evidence-Conditioned Synthesis:** LLMs generate candidate recipes containing only valid raw material tokens, explicitly prohibited from imputing unmeasured properties.

4. **Deterministic Validation Boundary:** Enforces the 6 automated safety checks. Flags unmeasured empirical properties as `NOT_VERIFIED` / `UNKNOWN`. If validation fails after 2 retries, the pipeline terminates and displays diagnostic logs; unconditional fallback to historical recipes is prohibited.
5. **Chemist Review & Sign-Off:** Chemist inspects ingredient provenance, adjusts active ratios, and provides authenticated digital authorization. Autonomous equipment booking is strictly locked out.

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4. Evidence Strategy & Data Contracts

4.1 Data Contracts (Pydantic v2)

```
from decimal import Decimal
from typing import List, Optional, Tuple
from pydantic import BaseModel, Field

class TargetSpecification(BaseModel):
    project_code: str; target_substrate: str; silicone_allowed: bool = False
    solids_pct_range: Tuple[Decimal, Decimal] = (Decimal("6.0"), Decimal("8.0"))
    ph_range: Tuple[Decimal, Decimal] = (Decimal("3.5"), Decimal("4.5"))
    viscosity_cp_range: Tuple[Decimal, Decimal] = (Decimal("50.0"), Decimal("300.0"))
    max_cost_inr_per_kg: Decimal = Decimal("22.0")
    min_softness_panel_score: Decimal = Decimal("4.0")  # Sensory panel scale 1–5
    target_stability_weeks_at_45c: int = 4              # 4w @ 45C: no separation/yellowing

class FormulationComponent(BaseModel):
    raw_token: str; resolved_rm_code: Optional[str] = None
    percentage: Decimal; effective_price_inr_per_kg: Decimal
    cost_contribution_inr: Decimal; is_ambiguous_synonym: bool = False

class CandidateFormulation(BaseModel):
    candidate_id: str; recipe: List[FormulationComponent]
    total_percentage: Decimal; total_rm_cost_inr: Decimal
    estimated_solids_pct: Decimal  # Authoritative solids mass balance (Decimal)
    estimated_ph: float            # Heuristic estimate (float, QSAR out of scope)
    estimated_viscosity_cp: float  # Heuristic estimate (float, bench validation needed)
    # Empirical properties (require physical testing; never LLM-hallucinated):
    stability_45c_4w_status: str = "NOT_VERIFIED"  # "PASS", "FAIL", "NOT_VERIFIED" (no separation/yellowing)
    softness_panel_score: Optional[Decimal] = None  # Scale 1–5; None/UNKNOWN if unmeasured
    evidence_citations: List[str]; compliance_flags: List[str]
    is_approved_by_chemist: bool = False
    approving_chemist_id: Optional[str] = None
```

Data Type & Empirical Property Architecture: All stoichiometry, active solids, formulation percentages, raw material prices, cost contributions, total recipe costs, and mass-balance checks operate strictly via Python `Decimal` fixed-point math without intermediate binary float rounding. Non-authoritative heuristic property estimates (`estimated_ph` , `estimated_viscosity_cp`) remain typed as `float` . Empirical properties (`stability_45c_4w_status` , `softness_panel_score`) require physical incubator and sensory panel test data; missing or incomplete measurements remain explicitly `UNKNOWN` or `NOT_VERIFIED` and cannot be claimed passed by generative text.

4.2 Retrieval Strategy: Relational SQL + BM25 vs. Vector DB

The draft proposal's nightly vector database is rejected for Kestrel's workload:

- **Small Data Corpus & Exact Filtering:** 28 raw materials and 12 project softener trials require strict numerical interval filtering (viscosity in [50, 300] cP, pH in [3.5, 4.5]) where vector embeddings degrade.
- **Relational Integrity:** Queries require exact joins across physical incubator logs (`Stability_Log`) and monthly pricing feeds (`Price_Feed`).
- **Selected Architecture:** Relational **PostgreSQL/SQLite** tables with **BM25 full-text indexing** over regulatory notes and chemical literature.

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5. Validation, Safety & Governance Gates

Gate	Validation Scope	Rule & Tolerance	Failure Mode Action
Gate 1: Mass Balance	Active solids & total formula sum	Σ w_i = 100.0% ± 0.05% (fixed-point `Decimal`)	Rejects malformed formulations (e.g. `T-09` 103.0% sum).
Gate 2: Ingredient Resolution	Chemical identity & supplier code	Primary key `RM-xxx`; flag synonym ambiguities	Disambiguates `RM-101` vs `RM-102` (`TEA-esterquat` in `T-04`).
Gate 3: Hard Constraints	Regulated & prohibited compounds	Silicone = 0.0% (strict ban); Isopropanol ≤ 1.0%	Hard block on prohibited compounds; advisory warnings.
Gate 4: Cost Normalization	Commercial pricing & packaging units	Active `Price_Feed` join on `RM-xxx`; FX ₹84.00/USD	Normalizes drum units (`RM-170` ₹32,000/200kg = ₹160/kg).
Gate 5: Evidence Trust	Historical trial & empirical validation	Ingest incubator logs & sensory panel records	Quarantines unapproved (`T-04`), phase-separated/yellowed (`T-05`), incomplete (`T-11`), and unmeasured softness cases as `NOT_VERIFIED` / `UNKNOWN`.
Gate 6: Chemist Sign-Off	Human governance barrier	Authenticated role-based digital signature	Mandatory chemist authorization; zero automated lab booking.

1. **Mass Balance Gate:** Enforces Σ w_i = 100.0% ± 0.05% using fixed-point `Decimal` math; rejects malformed formulations (`T-09` 103.0%).
2. **Ingredient Resolution Guard:** Maps raw tokens to master IDs; flags ambiguous synonyms (`RM-101` Supplier A vs `RM-102` Supplier B for `TEA-esterquat`).
3. **Hard Constraint Filter:** Blocks recipes with prohibited compounds (silicone = 0.0%); raises warnings on advisory limits (`RM-131` isopropanol ≤ 1.0%).
4. **Deterministic Costing Engine:** Evaluates active prices from `Price_Feed`, normalizing packaging units (₹32,000 / 200 kg = ₹160/kg for `RM-170` on drum basis) and currency ($1 USD = ₹84.00 historical basis); rejects unverified static scalar claims (`Summary_Apr` ₹16.90/kg).
5. **Evidence Trust & Empirical Quarantine:** The assistant is strictly prohibited from claiming thermal stability or sensory softness without supporting empirical test records. Stability requires verified 4-week 45°C logs with zero phase separation and zero yellowing; incomplete (`T-11`) or contradicted (`T-05`) trials are quarantined as `NOT_VERIFIED`. Softness requires verified panel testing (≥ 4/5); missing panel scores remain `UNKNOWN`. Generative imputation of physical or sensory pass labels is forbidden.
6. **Chemist Sign-Off Gate:** State transition to `APPROVED_FOR_BENCH` requires authenticated role-based digital signature; autonomous lab instrument scheduling is strictly prohibited.

6. Integration, Security & Production Operations

6.1 Proposed System Interfaces

- **LabTrack LIMS Interface (Proposed):** REST API / Read-Only Replica. Reads historical trials and equipment schedules; emits approved draft protocols for bench execution.
- **PurchaseOne ERP Interface (Proposed):** Read-only webhook / daily export. Pulls active supplier price feeds and currency exchange rates.
- **Literature Gateway (Proposed):** Caching wrapper over Crossref / arXiv / PubMed APIs for textile chemistry research.

6.2 Security, Governance & Role-Based Access Control

- **Role-Based Permissions:**
  - *Bench Chemist:* Submit briefs, inspect evidence, modify recipe proportions, sign off trial plans.
  - *Lead Chemist / Lab Manager:* Update advisory tolerances, resolve alias ambiguities, adjust list prices.
  - *System Administrator:* Manage API credentials, inspect audit trails, monitor telemetry.
  - *Strict Governance Boundaries:* No user role (including Lab Manager or Admin) can bypass prohibited chemical exclusions (silicone ban), unresolved ingredient collisions, invalid mass balance, or mandatory chemist sign-off.
- **Data Protection & Privacy (Proposed):** Enterprise Zero Data Retention agreements with LLM vendors. Encrypted in transit (TLS 1.3) and at rest (AES-256).

6.3 Observability, Telemetry & Audit Trail

- **Tamper-Evident Session Log:** Records brief inputs, retrieved trial IDs, candidate recipes, deterministic validation metrics, active configuration version hashes, and approving chemist ID in an append-only relational audit table with SHA-256 session integrity checks.
- **Telemetry & Drift Alerting:** Prometheus + Grafana dashboard tracking request latency, token consumption, and validation rejection rates. Triggers alert if rejection rate exceeds 15% in a rolling 1-hour window.

6.4 Operational Rollback & Incident Recovery Protocol

When production anomalies or safety regressions occur, the platform follows an explicit 6-step recovery protocol:

1. **Detection:** Automated monitors detect safety-barrier breaches (e.g. mass-balance breach > 0.05%, silicone leakage, corrupted price/raw-material ingestion), severe model extraction regression, or application faults.

2. **Immediate Containment (Circuit Breaker):** If deterministic safety checks or pricing/evidence data cannot be trusted, the orchestrator immediately **shuts down candidate generation** and places the system in a **safe, read-only state**. Chemists can query historical records, but new recipe drafting and export endpoints are hard-blocked.
3. **Version Reversion:** Application and model gateway configurations revert to the last known-good immutable release tag; data indices roll back to the previous verified relational snapshot and BM25 index.
4. **Evidence & Audit Preservation:** Immutable audit logs, failed inputs, validation diagnostics, active model/data version hashes, and incident timestamps are preserved in tamper-evident storage for root-cause analysis.
5. **Engineering Verification:** Candidate generation remains disabled until engineering executes the full golden regression suite (REG-01 through REG-11) and achieves 100% deterministic pass in staging.
6. **Governance Review & Re-Activation:** Normal operation resumes only after joint sign-off by Lead Systems Engineer and Lead Formulation Chemist. Rolling back a model or configuration **can never override or bypass deterministic validation gates or mandatory chemist approval**.

Incident Trigger Matrix:

- **Immediate Generation Shutdown (Circuit Breaker):** Triggered by deterministic barrier bypass, prohibited chemical leakage, unverified data corruption in `Price_Feed` / `RM_Master`, or mass-balance failure. Candidate generation halted immediately.
- **Warning & Ordinary Rollback:** Triggered by elevated latency (p95 > 15s), minor prompt extraction schema retry spikes, or non-breaking UI defects. Addressed via standard blue-green rollback without disabling read-only workspace functions.

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7. Cost Analysis & 8-Week Implementation Roadmap

7.1 Operating Cost Model (8 Chemists) & Target Reconciliation

Model pricing reflects published cloud developer benchmarks. Baseline workload assumes structured customer formulation briefs.

Cost Parameter	Option A: Tiered Routing (Recommended)	Option B: Un-Tiered Frontier Model
Pipeline Step 1 (Intake)	GPT-4o-mini / Haiku ($0.15/$0.60 per M) → **$0.00041/req**	Frontier ($3.00/$15.00 per M) → **$0.01200/req**
Pipeline Step 2 & 4 (Rules)	Python Deterministic Code → **$0.00000/req**	Python Deterministic Code → **$0.00000/req**
Pipeline Step 3 (Synthesis)	Prompt-Cached GPT-4o / Haiku → **$0.01175/req**	Un-cached Frontier Model → **$0.02250/req**
Composite Request Cost	**$0.01216 ≈ $0.0122 / request**	**$0.03450 / request**
Monthly Cost (352 req/mo)	**352 requests/month × $0.0122/request ≈ $4.29/month**	352 requests/month × $0.0345/request ≈ $12.14/month
Cost per Chemist / Month	**$4.29 ÷ 8 ≈ $0.537 / chemist/month**	$12.14 ÷ 8 ≈ $1.52 / chemist/month
Baseline Budget Status	**SATISFIED (Meets < $1.00/chemist/month target; 46.3% margin)**	**FAILED ($0.52 over budget ceiling; +52% overrun)**
Higher-Volume Scenario (1,760 req)	**1,760 requests/month × $0.0122/request ≈ $21.47/month**	1,760 requests/month × $0.0345/request ≈ $60.72/month
Higher-Volume Cost / Chemist	**$21.47 ÷ 8 ≈ $2.68 / chemist/month**	$60.72 ÷ 8 ≈ $7.59 / chemist/month
Higher-Volume Budget Status	**EXCEEDS BUDGET TARGET ($2.68 vs $1.00 target; +168% overrun)**	**EXCEEDS BUDGET TARGET ($7.59 vs $1.00 target; +659% overrun)**
Departmental Infrastructure (Hosting)	AWS ECS / DigitalOcean container + DB: **≈ $60–$100 / month**	Same infrastructure: **≈ $60–$100 / month**
Total Departmental Spend (Baseline + Infra)	**≈ $65–$105 / month total departmental spend**	**≈ $72–$160 / month total departmental spend**

- **Cost Assumptions & Estimates:** Token counts (Intake: ~800 in / 350 out; Synthesis: ~3,500 in / 1,200 out), model unit prices, request volumes (352 baseline vs 1,760 higher-volume), and prompt caching hit rates are engineering estimates rather than measured production figures.
- **Separation of API vs. Infrastructure:** Variable LLM API costs ($4.29/month baseline) are tracked separately from fixed cloud infrastructure ($60–$100/month hosting & database).
- **Budget Target Reconciliation:** The baseline workload of 352 requests/month (44 requests/chemist/month across 8 chemists) achieves $0.537/chemist/month, successfully satisfying the departmental < $1.00/chemist/month API budget target. The higher-volume scenario of 1,760 requests/month (220 requests/chemist/month) reaches $2.68/chemist/month, which **exceeds the $1.00/chemist/month budget ceiling** and requires query rate-limiting or prompt token caps if sustained.

7.2 8-Week Phased Implementation Roadmap (Team of 3 Engineers)

Sprint	Timeline	Key Engineering Deliverables	Milestones & Success Criteria
Sprint 1	Weeks 1–2	Relational schema migration (12 sheets), Python deterministic engine (`Decimal` stoichiometry, UOM/FX converters).	Zero mass-balance leaks; 100% pass on historical integrity suite.
Sprint 2	Weeks 3–4	Request intake parser, SQL/BM25 retrieval engine, tiered LLM gateway with constrained Pydantic output schemas.	Validated JSON extraction; bidirectional retrieval operational.
Sprint 3	Weeks 5–6	Chemist web workspace, candidate comparison UI, append-only audit logging, authenticated sign-off workflow.	Complete end-to-end flow; zero automated lab bookings.
Sprint 4	Weeks 7–8	Week 7: Shadow pilot with 2 senior chemists (Lab 1) & formal exit-gate review. Week 8: Target rollout across all 8 chemists ONLY IF all exit criteria pass; otherwise dedicated to remediation, regression testing & retesting.	Target schedule contingent on exit gate; 100% exit criteria met & 2-chemist sign-off before wider rollout; unresolved critical defects defer rollout (safety never bypassed for schedule).

Sprint 4 Pilot Exit Gate & Remediation Protocol

The 8-week implementation plan serves as a **target engineering schedule**, not an unconditional rollout deadline. Organization-wide rollout to all 8 chemists across both laboratories is permitted only after every documented acceptance criterion passes and both pilot senior chemists provide documented sign-off:

1. **Critical Safety & Prohibited Ingredients:** 100% pass rate on silicone exclusion and advisory threshold regression suites (`RM-140` block, `RM-131` warning).
2. **Deterministic Calculation Integrity:** 100% pass rate on `Decimal` mass-balance (100.0% ± 0.05%), unit conversion, and costing reference benchmarks.
3. **Evidence Integrity & Quarantine:** Missing (`T-11`), ambiguous (`RM-101/102`), and contradictory (`T-05`) evidence correctly flagged and quarantined rather than silently accepted.
4. **Empirical Property Grounding:** 100% pass rate on stability (4w @ 45°C: zero phase separation/yellowing) and softness (panel score ≥ 4/5) regression suites; unverified or incomplete records quarantined as `UNKNOWN` / `NOT_VERIFIED` .
5. **Chemist Governance Enforcement:** Mandatory authenticated role-based digital signature enforced on every recommendation; zero autonomous lab bookings emitted.
6. **Pilot User Acceptance Sign-Off:** Both pilot senior chemists formally review and sign off on recommendation quality and UI workflow usability; unresolved critical defects block wider rollout.

Remediation, Retest & Deferral Policy: Week 7 is strictly dedicated to the 2-chemist shadow pilot and formal exit-gate evaluation. Week 8 is held available for remediation, regression testing, and retesting against the 74-test regression suite (spanning the 11 named golden regression scenarios, REG-01 through REG-11) if any acceptance criterion fails or critical defects emerge. Rollout to all 8 chemists proceeds if and only if remediation achieves 100% exit-gate compliance. If critical defects remain unresolved at the end of Week 8, organization-wide rollout is deferred until remediation and retesting are complete. Safety checks, validation gates, and chemist approval requirements will never be waived or bypassed to meet calendar deadlines.

8. Summary of Engineering Rationale

By establishing a **deterministic safety barrier around the generative model**, the Kestrel assistant captures the productivity benefits of AI while eliminating the operational hazards of hallucinated stoichiometry, unnormalized packaging costs, unapproved trial scale-ups, and unauthorized lab bookings. This delivers an auditable, cost-effective ($0.537/chemist/month baseline) platform built for practical R&D bench adoption.